Related Rates and Optimization: Diagnosing the Setup

AP Calculus AB / BC — Applications of Derivatives

Setup discipline — the four errors this sheet targets:

- **Substituting too early.** Instantaneous values go in *after* differentiating. A number written into the relation first is treated as a constant and its rate disappears.
- **Leaving a second variable in.** Use the geometry to eliminate variables until one remains, then differentiate.
- **Inverting a ratio.** Solve the differentiated equation symbolically for the rate you want before any arithmetic.
- **Writing the wrong constraint.** An optimization constraint must describe the actual situation, not the familiar formula that resembles it.

For every find-and-fix item, quote the specific line of the setup that is wrong, then redo the problem from that line on.

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1. **Setup step.** A circle's area is $A = \pi r^2$. Differentiate with respect to r to get the factor that will multiply $\dfrac{dr}{dt}$ when the chain rule is applied.

Answer: ____________________

2. **Setup step — eliminate first.** A cone's volume is $V = \frac{1}{3}\pi r^2 h$, and for this cone the height is always twice the radius. Substitute the constraint so V depends on r alone, then differentiate with respect to r.

Answer: ____________________

3. A spherical balloon is inflated at a constant 100 cubic centimetres per second. Find $\dfrac{dr}{dt}$ at the instant when the radius is 5 centimetres. Round to three decimal places.
Write the differentiated relation before you substitute anything.

Answer: ________________

4. A 10-foot ladder leans against a vertical wall. Its base slides away from the wall at 2 feet per second. Find the rate at which the top of the ladder slides down at the instant the base is 6 feet from the wall. Round to three decimal places.

Answer: ________________

5. Find and fix the mistake. A conical tank has radius 3 metres at the top and depth 6 metres, and water drains out at 12 cubic metres per minute. A student writes $\frac{dV}{dt} = \frac{1}{3}\pi r^2 \frac{dh}{dt}$, treating the radius as the fixed value 3.

Name what that step assumes about the water's surface, then find $\frac{dh}{dt}$ correctly when the depth is 4 metres. Round to three decimal places.

Answer: ________________

6. Find and fix the mistake. A rectangular pen is built against a barn wall, so fencing is needed on only three sides. There are 200 feet of fencing. A student writes the constraint $2x + 2y = 200$.

Correct the constraint, write the area as a function of x alone, differentiate it, and find the x that maximises the area.

Answer: ________________

7. Open-box optimisation. An open-top box is made from a square sheet measuring 24 inches by 24 inches by cutting a square of side x from each corner and folding up the sides. Write the volume as a function of x, differentiate it, and find every critical value. State which critical value must be rejected and why.

Answer: ________________

8. Explain. A classmate substitutes the given instantaneous radius into the volume formula before differentiating, gets a constant, and concludes the rate is zero. Write a short paragraph explaining what went wrong, distinguishing a quantity that varies from one that is genuinely constant, and stating exactly when substituting a given value is legal.